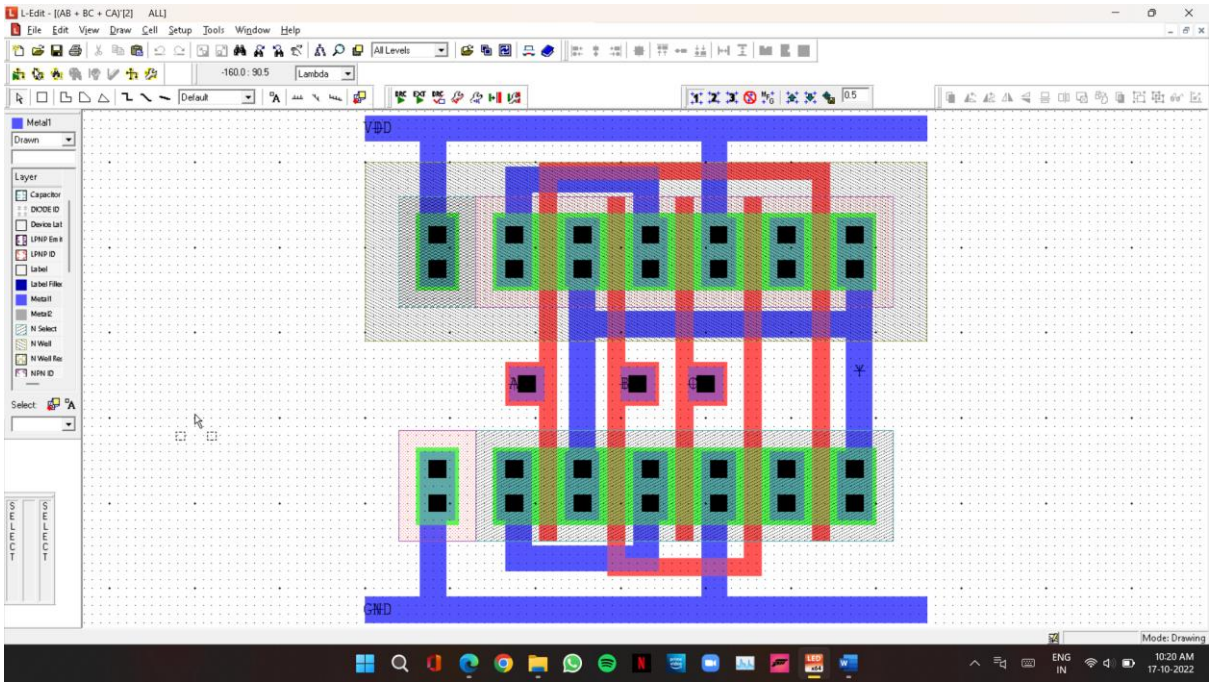
LAB – 11

(AB + BC + CA)'

K.N Sujay 121910402035

Outputs:



T-Spice - [(AB + BC + CA)']

File Edit View Simulation Table Setup Window Help

```
*****************************************
.MODEL NMOS NMOS LEVEL=1 VTO=1
.MODEL PMOS PMOS LEVEL=1 VTO=-1
M1 Y A 1 GND NMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (-118.5 70.5 -116.
M2 1 B Y GND NMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (-93.5 70.5 -91.5
M3 GND C 1 GND NMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (-59.5 70.5 -57.
M4 2 B GND GND NMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (-16.5 70.5 -14.
M5 Y A 2 GND NMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (17.5 70.5 19.5 79
M6 Y A 3 VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (-118.5 106.5 -116
M7 3 B Y VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (-93.5 106.5 -91.5
M8 VDD C 3 VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (-59.5 106.5 -57
M9 4 B VDD VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (-16.5 106.5 -14
M10 Y A 4 VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (17.5 106.5 19.5
VDD VDD GND DC 5
*VG A GND DC 5
*VG1 B GND DC 5
*VG2 C GND DC 5
*.DC VG 0 5 0.01
```

| Status | Input file | Start Time/D... | Elaps... |
|---|---|---|---|
| finished | (AB + BC + CA)'.spc | 20:50:51 Oc... | 00:0... |
| finished | (AB + BC + CA)'.spc | 20:52:07 Oc... | 00:0... |
| finished | (AB + BC + CA)'.spc | 20:53:27 Oc... | 00:0... |

Ready Ln 38, Col 11

ENG IN 08:54 PM 16-10-2022

T-Spice - [(AB + BC + CA)']

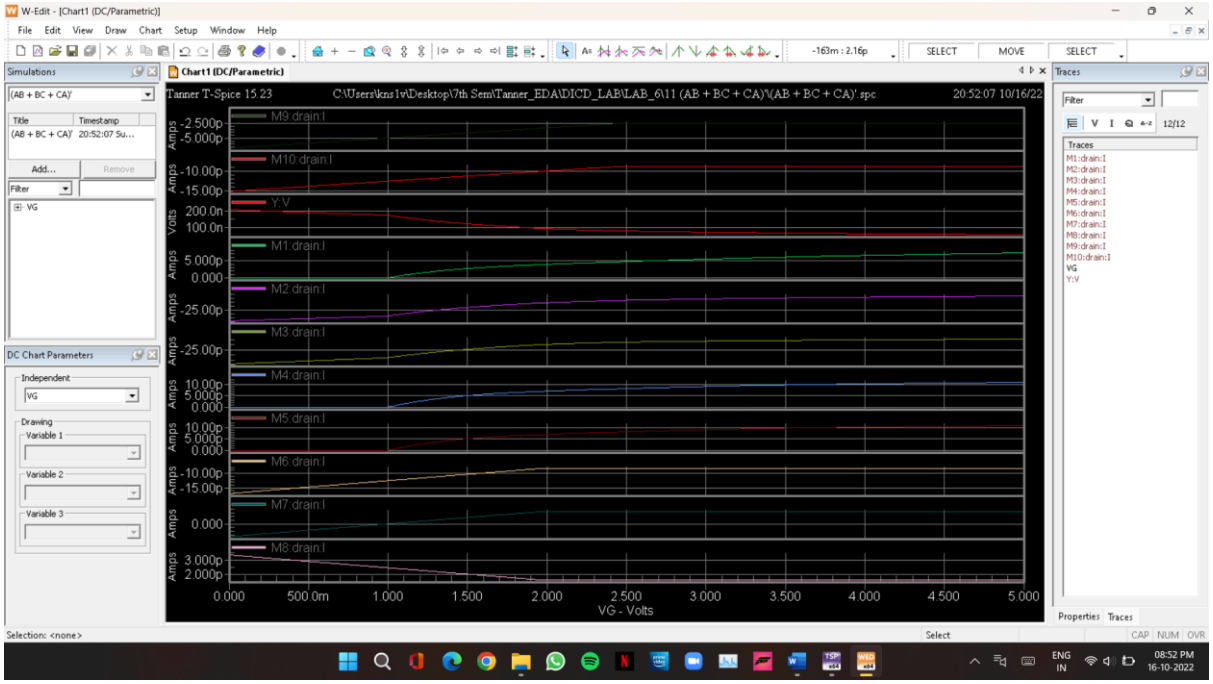
File Edit View Simulation Table Setup Window Help

```
M3 GND C 1 GND NMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (-59.5 70.5 -57.
M4 2 B GND GND NMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (-16.5 70.5 -14.
M5 Y A 2 GND NMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (17.5 70.5 19.5 79
M6 Y A 3 VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (-118.5 106.5 -116
M7 3 B Y VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (-93.5 106.5 -91.5
M8 VDD C 3 VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (-59.5 106.5 -57
M9 4 B VDD VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (-16.5 106.5 -14
M10 Y A 4 VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (17.5 106.5 19.5
VDD VDD GND DC 5
*VG A GND DC 5
*VG1 B GND DC 5
*VG2 C GND DC 5
*.DC VG 0 5 0.01
*.PRINT V(Y) I(M1) I(M2) I(M3) I(M4) I(M5) I(M6) I(M7) I(M8) I(M9) I(M10)
VG A GND PULSE (0 5 0 1N 1N 10N 20N)
VG1 B GND PULSE (0 5 0 1N 1N 20N 40N)
VG2 C GND PULSE (0 5 0 1N 1N 40N 80N)
.TRAN 1N 400N
.PRINT V(Y) V(A) V(B)  V(C)
.END
```

| Status | Input file | Start Time/D... | Elaps... |
|---|---|---|---|
| finished | (AB + BC + CA)'.spc | 20:50:51 Oc... | 00:0... |
| finished | (AB + BC + CA)'.spc | 20:52:07 Oc... | 00:0... |
| finished | (AB + BC + CA)'.spc | 20:53:27 Oc... | 00:0... |

Ready Ln 38, Col 11

ENG IN 08:54 PM 16-10-2022


W-Edit - [Chart1 (DC/Parametric)]
Chart1 (DC/Parametric)
Tanner T-Spice 15.23
C:\Users\kns1v\Desktop\7th Sem\Tanner_EDA\DICD_LAB\LAB_6\11 (AB + BC + CA)\(AB + BC + CA)'.spc
20:52:07 10/16/22
M9:drain:I
M10:drain:I
Y:V
M1:drain:I
M2:drain:I
M3:drain:I
M4:drain:I
M5:drain:I
M6:drain:I
M7:drain:I
M8:drain:I
VG - Volts

W-Edit - [Chart1 (Transient)]
Chart1 (Transient)
Tanner T-Spice 15.23
C:\Users\kns1v\Desktop\7th Sem\Tanner_EDA\DICD_LAB\LAB_6\11 (AB + BC + CA)\(AB + BC + CA)'.spc
20:53:27 10/16/22
Y:V
C:V
B:V
A:V
Volts
Seconds